

Cramér–Wold Device: Graduate-Level Exercises

Twelve qualifying-exam style problems with hints only.

Problem 1. State and prove the Cramér–Wold device.

Hint: Show that convergence of all linear combinations determines the multivariate distribution.

Problem 2. Use the Cramér–Wold device to prove multivariate convergence of a sequence of Gaussian vectors.

Hint: Identify the limiting distribution of every $a^T X_n$.

Problem 3. Show that convergence of coordinate marginals alone does not imply joint convergence.

Hint: Construct a counterexample and compare with Cramér–Wold.

Problem 4. Prove the multivariate Central Limit Theorem from the univariate CLT using the Cramér–Wold device.

Hint: Apply the univariate CLT to $a^T X_n$.

Problem 5. Let $X_n = (X_{n1}, X_{n2})$. Show that if $a^T X_n$ converges in distribution for every a , then the limits are consistent with a unique multivariate law.

Hint: Use characteristic functions.

Problem 6. Apply the Cramér–Wold device to derive the asymptotic distribution of a vector of sample means.

Hint: Study arbitrary linear combinations.

Problem 7. Show how the Cramér–Wold device interacts with Slutsky's theorem.

Hint: Apply Slutsky to $a^T X_n$.

Problem 8. Use the Cramér–Wold device to prove joint asymptotic normality of two estimators.

Hint: Start from linear combinations $c_1 T_{1n} + c_2 T_{2n}$.

Problem 9. Give an example where tightness is needed in a multivariate convergence argument.

Hint: Compare convergence of projections with convergence in law.

Problem 10. Show that if $X_n \Rightarrow X$ in $\mathbb{R}^k$, then $a^T X_n \Rightarrow a^T X$ for every a .

Hint: Use the Continuous Mapping Theorem.

Problem 11. Prove the vector-valued Delta Method using the Cramér–Wold device.

Hint: Reduce to scalar linear combinations.

Problem 12. Use the Cramér–Wold device to establish asymptotic normality of a multinomial proportion vector.

Hint: Apply the device to arbitrary contrasts.